

DMP04 FAQs

1. How can I make sure that my algorithm will consider all exceptional situations?

Algorithms won't handle all exceptional situations on their own. We, as algo developers, need to code that checks for all foreseeable cases. We also need to ensure that the algorithm does not stop working when unexpected situations occur. The standard way to handle such situations is to use try/except blocks whenever possible.

2. How can I use my backtesting strategy and take it into a live trading setup?

Ideally, the trading algorithm should be the same, just change the data module/script/function/class to read the data, but the logic of the strategy ideally must be the same code in order to avoid introducing small differences. However, there are times when the backtesting program and the live trading (execution) program would be two different algos. First, we start with a research process where we try to come up with a trading hypothesis. Next, to validate it, we create a backtesting algorithm. Depending on its result, we make changes to the algorithm and arrive at a trading model. Then, we make/develop a live trading algorithm, aka execution algorithm. This step involves using a broker's API to connect your strategy with markets and perform trading in real-time.

3. How to interpret the gap between the accuracy between in-sample and out-sample data?

Ideally, the accuracy for the training dataset and the testing dataset should be comparable. If there are significant differences in the accuracy of both, then it is a matter of concern. It is a sign that our model is either underfit or overfit. We will cover this in detail in the MLT lectures while discussing the bias-variance tradeoff.

4. What is the difference between supervised and unsupervised learning?

Please refer to the following resources:

- <https://blog.quantinsti.com/machine-learning-basics/>
- <https://blog.quantinsti.com/trading-using-machine-learning-python/>
- <https://blog.quantinsti.com/machine-learning-classification/>
- <https://blog.quantinsti.com/scikit-learn-tutorial/>

5. What should be the size of the training dataset for stock prediction in ML?

There's no such generalization on the number of data points that we need for training purposes. It usually depends on the size of the dataset and the distribution of the target variable (in the case of supervised learning). Practitioners split in a ratio of 70-30, 75-25, or 80-20, depending on the task they are carrying out.

6. Which backtesting method(vectorized/event-driven) is more popular?

Both methods are equally used. The event-driven approach provides flexibility to include any event in the backtesting. On the other end, if you want to validate your idea quickly, a vectorized approach is the go-to approach.